

HETEROGENEOUS RESERVOIR ENSEMBLES DECODE DEEPER AND PREDICT WORSE THAN HOMOGENEOUS ONES AT MATCHED WIDTH

A P R E P R I N T

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ABSTRACT

An open-ended design brief became a controlled test of representational heterogeneity. Three arms at matched total width $N=640$, driven by an identical text stream: **S**, one intact reference reservoir; **M**, five independent same-family reference draws concatenated; **C**, five structurally *different* reservoir families concatenated — the reference sparse-tanh mixer, an addressed-memory reservoir with permutation recurrence, an 8-bit state-quantized reservoir, a periodic block-reset reservoir, and a block-diagonally fragmented reservoir — with **C** drawing the identical per-slot seeds as **M**, so that family identity is the only difference. Finding, replicated on a fresh hardware-drawn seed with zero reversals: **the heterogeneous ensemble predicts worse than both S and M on ridge and logistic validation bits per character (bpc), while decoding strictly deeper than both** — a complete three-way rank inversion (bpc ranking $M < S < C$; depth ranking $C > S > M$), reproduced exactly. The C–M gap is stable across seeds to roughly 0.005–0.007 bits (ridge 0.0577/0.0532, logistic 0.0625/0.0560). A companion result: **M** lands within 0.0175 bits of **S** or closer on every instrument at both seeds, extending the program's fragmentation-is-nearly-free findings to five-way at a new width. This is the program's fourth independent route to the decodability–usability gap — and the first produced by cross-family heterogeneity itself rather than a single mechanism inside one family.

Keywords reservoir computing · echo state networks · heterogeneous ensembles · decodability vs. usability · rank inversion · pre-registration

§1 The question

Combining structurally different reservoirs is a long-standing idea — decoupled sub-reservoirs were proposed as an architecture almost two decades ago [2] — and the intuitive expectation is that diversity should help: five kinds of memory ought to cover for one another's blind spots. This program has repeatedly measured a gap, however, between what a state *stores* and what a trained readout can *use*. The registered question therefore became: at strictly matched total width, does concatenating five families buy usable prediction, or does it buy something else?

§2 Method

All arms at N=640 total width, with the program's standard parameters throughout (spectral radius $\rho=0.95$, leak 1.0, bias 0.2, tanh activation where applicable [1]), standard text8 splits [3], 2M training characters:

S: one intact reference reservoir, N=640
 M: 5 x reference(N=128), independent seeds {0..4}, concatenated
 C: 5 x N=128, SAME per-slot seeds as M, different families:
 slot 0 reference sparse-tanh mixer
 slot 1 addressed-memory ESN (permutation + Hadamard coding)
 slot 2 state-quantized ESN, b=8 bits per unit
 slot 3 periodic block-reset schedule, 8 blocks
 slot 4 block-diagonal fragmentation, H=8

M and C sharing seed draws slot-for-slot isolates the causal question — does family identity matter? — from a random-draw confound. Construction code for each component reservoir was copied verbatim from its source study. The five component families are drawn from earlier work in this program: the addressed-memory ESN, the periodic block-reset reservoir, and the block-fragmented reservoir have their own whitepapers; the state-quantized family comes from the program's quantization runs.

Instruments: ridge and logistic validation bpc, U3 decode depth — the linear decode-depth probe, so designated in the program's internal instrument numbering. The replication seed (3643503230) was hardware-drawn and recorded before running.

§3 Pre-registered predictions

A disclosed pilot run showed the direction before any prediction was written — and it was *not* the direction the brief's premise suggested. The registered predictions followed the pilot, with that premise retained as the named alternative:

- **CC1/CC2:** C predicts worse than both S and M on ridge (2/2) and logistic (2/2) validation bpc. Named alternative (the brief's premise, and a registered surprise if it occurred): $C \leq M$ — the benefit of diversity reasserts itself at 20× the pilot's data.
- **CC3:** C decodes deeper than both S and M (2/2).
- **CC4 (headline):** the full rank inversion — bpc best-to-worst M, S, C on both instruments; depth deepest-to-shallowest C, S, M. Hits only if CC1–CC3 all hit together. Named alternative: any partial inversion.
- **CC5 (supporting):** M within 0.03 bits of S on both instruments — five-way same-family fragmentation is nearly free.
- **Bands:** C's logistic val in [2.85, 3.10]; the C–M logistic gap ≥ 0.01 bits; the C–M depth gap ≥ 0.2 characters.

The replication re-registered all five ordinals (R-CC1–R-CC5, with a widened ± 0.05 tie band for R-CC5) and both surviving bands, with promotion pre-committed to R-CC1 through R-CC4 hitting clean.

§4 Results

Table 1. Both seeds (run ledger #33 / #36; values given as seed 0 / seed 3643503230). All gates pass, no voided logistic fits, zero reversals across every registered comparison.

arm	ridge val bpc	logistic val bpc	U3 depth
S – intact reference reservoir	3.2652 / 3.2580	2.7843 / 2.7787	9.28 / 9.33
M – five same-family draws	3.2477 / 3.2499	2.7683 / 2.7708	8.97 / 9.03
C – five different families	3.3054 / 3.3031	2.8308 / 2.8268	9.49 / 9.52

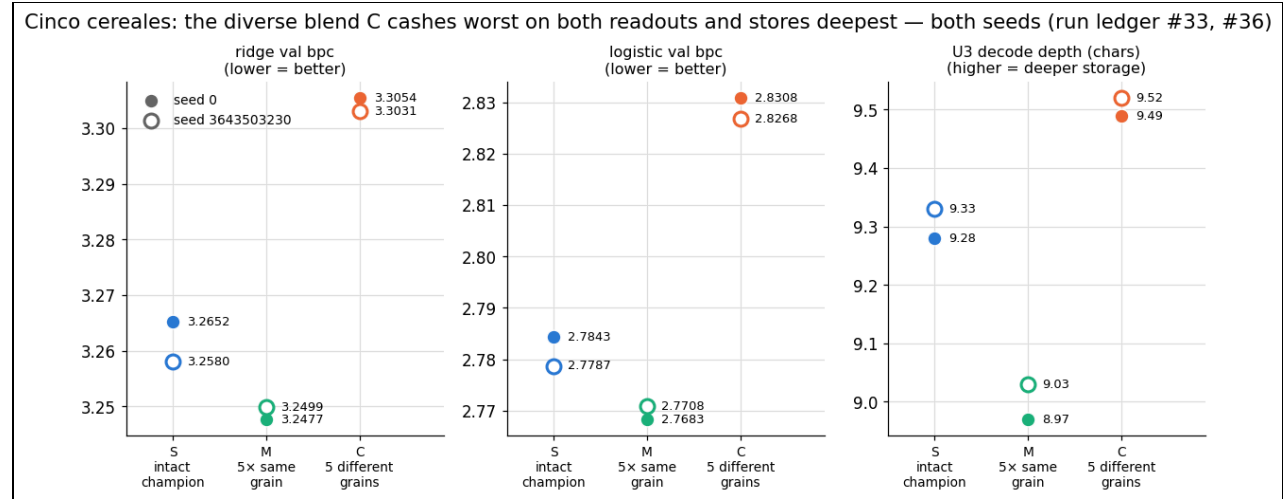


Figure 1. The rank inversion, both seeds (run ledger #33, #36). Filled markers: seed 0; open: seed 3643503230. On both trained-readout instruments the ordering is $M < S < C$ (best to worst); on decode depth it is exactly reversed, $C > S > M$.

All primary ordinals hit at both seeds, with zero reversals. C predicts worst on both instruments and decodes deepest, at both seeds; the full three-way rank inversion (CC4, R-CC4) reproduces exactly. The effect's size is stable, not only its direction: C–M gaps of 0.0577/0.0625 bits (ridge/logistic) at seed 0 against 0.0532/0.0560 at the replication seed, and C–M depth gaps of 0.52 and 0.49 characters.

Five same-family draws are nearly free — and slightly better than one intact reservoir. M lands within 0.0175/ 0.0160 bits of S at seed 0 and 0.0081/0.0079 at the replication seed, and is directionally *better* than S on both instruments at both seeds — extending the two-way (orthogonal-reservoir) and 64-way (block-diagonal fragmentation) results to five-way at a new width.

Bands: 2/3 at seed 0, with the miss in the better-than-predicted direction. C's logistic validation bpc (2.8308) landed below the registered [2.85, 3.10] floor — the heterogeneous ensemble predicted *better* than the pilot-informed band anticipated while still being the worst arm — matching this program's established pattern of quantitative bands missing toward better-than-predicted performance. The gap bands held at both seeds.

§5 Discussion

Heterogeneity itself trades usability for decodability. The program's three prior routes to this dissociation each ran through a single mechanism inside one family — an addressing scheme, a future-keyed linear bridge, a quantization dial. Here the same signature emerges from *combining unrelated substrates*, with no single mechanism doing the work: five kinds of memory, concatenated, retain more of the recent stream than any homogeneous arm at the same width, and a trained readout converts strictly less of it into prediction. Diverse representations are harder to use than they are to fill.

What this does not establish. Whether the predictive-capture law (x_c) prices this heterogeneous state the same way it priced the other three dissociations is untested — recorded in the program's work queue as answerable from result files already on disk. Width and budget are fixed at $N=640$ and 2M characters; nothing here shows that the inversion survives other operating points. Uncertainty is reported as a min–max range across two seeds, and the program's independent audit of this entry is still pending.

§6 References

1. H. Jaeger, "The 'echo state' approach to analysing and training recurrent neural networks," GMD Report 148, German National Research Center for Information Technology, 2001.
2. Y. Xue, L. Yang, S. Haykin, "Decoupled echo state networks with lateral inhibition," *Neural Networks* 20(3), 2007.
3. M. Mahoney, "Large text compression benchmark" (text8), mattmahoney.net/dc/textdata.

§7 Provenance

- **ef80c874** — 2026-08-12-caity-cinco-cereales: the initiating design brief; 3 arms, seed 0 (run ledger #33). Pilot disclosed before registration; CC1–CC5 all hit $\Rightarrow$ provisional, replication auto-queued, an intake record filed on the possible connection to the predictive-capture law.
- **2251de15** — 2026-08-12-cinco-cereales-rerun1: seed 3643503230, hardware-drawn with the raw draw recorded before running (run ledger #36). All five ordinals and both bands hit, zero reversals $\Rightarrow$ promoted per the pre-stated rule.

All claims judged strictly against the registered wording, the band miss included above; 2 seeds; replicated before publication; the program's independent audit of this entry is still pending. Internal designation: cinco cereales.

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